Description

Illustration





Figure caption

1. Decorative facing brick. 2. Wall clad in facing brick. Images used under license from Shutterstock.com

The material

Brick is as old as Babylon (4000 BC) and as durable. Palaces and temples of Mesopotamia were built of stone and sun-dried brick. It is the most ancient of all man-made building materials and belongs to that equally ancient building method of masonry construction. The regularity and proportions of bricks makes them easy to lay in a variety of patterns, and their durability and relative low cost makes them an ideal material for the construction of buildings. Clay - the raw material from which bricks are made - is available almost everywhere; finding the energy to fire them can be more of a problem. Pure clay is gray-white in color; the red color of most bricks comes from impurities of iron oxide. Yellow, gray and even black bricks can be made by varying the proportion of various clays and other ingredients and carefully controlling the heat of firing.

Brick is used mainly for walls and foundations. Common brick is used where appearance is not important, or where it will be plastered or rendered. Engineering brick is used where high strength, low porosity and high frost resistance are required. Facing brick is used for external work where good appearance is needed. Brick walls may be structural or non-structural; the latter being much more common as a veneer material spanning between the columns of a structural frame.

Facing brick is a kiln-fired brick used for external surfaces where good appearance and weathering are important. It is available in many shades (red, russet, orange, buff, cream, yellow, brown and blue) and textures (smooth, dragfaced, creased, rolled and rustic versions are available). The mechanical and thermal properties resemble those of engineering brick.

The quoted properties are those for a single brick rather than that of brickwork, in which the properties and adhesion of mortar would have to be taken into consideration (most obviously becoming a dominant factor when loading in tension).

Compositional summary

50-80% alumina and 20-35% silica

General properties

Density	2.05e3	-	2.3e3	kg/m^3

Price	* 0.747	-	1.99	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	4	-	8	GPa
Shear modulus	* 1.67	-	3.33	GPa
Bending modulus	* 4	-	8	GPa
Poisson's ratio	* 0.2	-	0.23	
Yield strength (elastic limit)	3	-	6	MPa
Tensile strength	3	-	6	MPa
Compressive strength	20	-	50	MPa
Bending strength	3	-	6	MPa
Elongation	* 0.04	-	0.15	% strain
Hardness - Vickers	8	-	30	HV
Fatigue strength at 10^7 cycles	* 1.2	-	2.4	MPa
Fracture toughness	* 0.00225	-	0.018	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.00875	-	0.0139	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	1.11	-	1.25	m.°C/W
Thermal expansion coefficient	5	-	8	µstrain/°C
Specific heat capacity	750	-	850	J/kg.°C
Melting point	1.2e3	-	1.5e3	°C
Maximum service temperature	673	-	877	°C
Flammability	Non-flammable			
Emissivity	0.75	-	0.93	

Hygro-thermal properties

Water absorption	2	-	7	%
Water vapor permeability	2.5e-12	-	2e-11	kg.m/s.m^2.Pa

Frost resistance	Good			

Electrical properties

Electrical conductor or insulator?	Good insulator			
Electrical resistivity	1e14	-	3e14	µohm.cm
Dielectric constant (relative permittivity)	4	-	10	
Dissipation factor (dielectric loss tangent)	0.001	-	0.01	

Optical properties

Natural color

Red or gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	Yes

Acoustic properties

Sound absorption	Poor
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Moldability	2	-	4
Machinability	1	-	2

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Acceptable
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Limited use
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.188	-	0.208	kg/kg
Embodied energy (virgin grade production)	* 3.27	-	3.61	MJ/kg
Water Usage	* 5.27	-	5.83	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 17.4	-	23.1	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.14	-	1.51	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 0.0188	-	0.0208	kg/kg
Embodied energy, recycling	* 0.458	-	0.505	MJ/kg
Recycle fraction in current supply	0			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Bricks and the mortar used to bond them are strong in compression but weak in tension. This largely determines the way they are used: brick structures (like those of stone) are designed in such a way that the loads are compressive everywhere, ruling out cantilevered or very slender forms. The face of brick can be molded and glazed, allowing great freedom of decoration, and the pattern and natural color variation gives a visually attractive surface. Low-fired, unglazed brick is vulnerable to water-penetration and degradation, and should be screened from direct weathering. High-fired or glazed brick is extremely durable. The durability of brick walls, as a building system, is dependent on the proper composition and tooling of the mortar joint and adequate weep holes that allow trapped liquid water to escape to the outside from inside the brick cavity. Freeze-thaw cycles, in which trapped water expands, are a common mechanism for cracked brick and spalled mortar.

The standard brick size is 3 1/2 in. (w) x 2 1/4" in. (d) x 8" in. (h), with relatively large tolerances. Modular bricks are sized such that three courses plus mortar joints amount to 8 inches in height. Modular, engineer modular, engineer standard, Roman, Norman, King size and other standard sizes are used for a variety of purposes and reasons. Non-standard sizes are also available. Solid, cored, hollow and 'frogged' bricks are all widely available.

Three grades of building and facing bricks are available in the US; Grade SW for use in conditions of severe weathering, MW for moderate weathering and NW for negligible weathering. Three types of facing brick are used; Type FBX (highly consistent dimensional and color uniformity), Type FBS (wide color variation and less dimensional consistency), and Type FBA (great color and dimensional variations).

Technical notes

Brick is fired clay: hydrous aluminum silicate, with impurities of potash, soda, lime, and oxides of iron. On heating to 900-1200C, the aluminum silicate reacts with soda to form a viscous glass - it is this process that bonds the brick together. Most brick is porous; if it is to resist water it is glazed. To do this the surface of the fired brick is painted with a mixture of glass-forming fluxes and the brick is re-fired, melting the glaze, and forming a glassy surface that can be colored.
Large pores and low porosity are the most important factor in establishing if a particular brick type will be durable.
The quoted properties are those for a single brick rather than that of brickwork, in which the properties and adhesion of mortar would have to be taken into consideration (most obviously becoming a dominant factor when loading in tension).

Typical uses

The external, visible surfaces of domestic and industrial buildings and walls.

Related standards

mortar: ASTM C144 (mortar sand), ASTM C150 (Portland cement), ASTM C207 (lime), ASTM C91 (masonry cements), ASTM C270 (mortar composition)
brick: ASTM C62, C216, C652 (grades and types of brick), ASTM C902 (paving bricks)
BS 3921 (1969)
CSI/CSC Masterformat Section Numbers:
04 05 16 Masonry Grouting
04 05 23 Masonry Accessories
04 20 00 UNIT MASONRY
04 21 00 Clay Unit Masonry
04 23 00 Glass Unit Masonry
04 50 00 REFRACTORY MASONRY
04 51 00 Flue Liner Masonry
04 57 00 Masonry Fireplaces

Links

ProcessUniverse
Producers
Reference

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.